MAT 1013: Discrete Mathematics
Session 12 : Functions

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Topics

- Functions as special types of relations
- Domain, codomain, and range
- Function notation and evaluation
- Representations of functions using ordered pairs, arrow diagrams, tables, formulas, and graphs
- Equality of functions
- Injective, surjective, and bijective functions
- Constant functions and identity functions
- Composition of functions
- Associativity of function composition
- Inverse relations and inverse functions
- Conditions for the existence of an inverse function
- Images and preimages of sets
- Properties of images and preimages
- Real-valued functions and their graphs
- Natural domain and range of real functions
- Special Properties and Types of functions
- Transformations of graphs
- Graphs of inverse functions
- Standard functions, their graphs, and their properties
- Piecewise-defined functions

Learning Outcomes

At the end of this lesson, students should be able to:

1. explain functions as a special type of relation;
2. identify the domain, codomain, and range of a function;
3. represent functions using ordered pairs, arrow diagrams, tables, formulas, and graphs;
4. determine whether a function is injective, surjective, or bijective;
5. form compositions and determine the domain of a composite function;
6. determine whether a function has an inverse and calculate it when it exists;
7. calculate images and preimages of sets;
8. count functions, injections, and bijections between finite sets;
9. interpret the graphs and principal properties of real functions;
10. sketch transformations of standard graphs.

1 From Relations to Functions

Recall that a relation from a set A to a set B is any subset of the Cartesian product $A \times B$. A function is a relation in which every element of the first set is assigned exactly one element of the second set.

Definition

Let A and B be nonempty sets. A *function* f from A to B is a relation $f \subseteq A \times B$ such that:

1. for every $x \in A$, there exists $y \in B$ such that $(x, y) \in f$;
2. if $(x, y_1) \in f$ and $(x, y_2) \in f$, then $y_1 = y_2$.

We write

$$f : A \to B,$$

and write $f(x) = y$ when $(x, y) \in f$.

The first condition says that every input must have an output. The second condition says that an input cannot have two different outputs.

Domain, Codomain, and Range

For a function $f : A \to B$:

- A is the *domain*;
- B is the *codomain*;
- the *range* or *image* of f is
$$f(A) = \{f(x) : x \in A\} \subseteq B.$$

Remark

The codomain is part of the definition of the function. The range consists only of the values that the function actually takes. Therefore, the range may be smaller than the codomain.

Example 1

Let $A = \{1, 2, 3, 4\}$ $B = \{a, b, c, d\}$ and let $f = \{(1, a), (2, b), (3, b), (4, c)\}$.

Then $f : A \to B$ is a function, and

$$\mathrm{Dom}(f) = A, \qquad \text{codomain} = B, \qquad \mathrm{Ran}(f) = \{a, b, c\}.$$

Activity 1: Function or Not?

For each relation below, determine whether it defines a function from A to B. Give a reason.

1. $A = \{1, 2, 3\}$, $B = \{a, b\}$, and $R = \{(1, a), (2, b), (3, a)\}$.
2. $A = \{1, 2, 3\}$, $B = \{a, b\}$, and $S = \{(1, a), (1, b), (2, a), (3, b)\}$.
3. $A = \{1, 2, 3\}$, $B = \{a, b\}$, and $T = \{(1, a), (2, b)\}$

Representations of a Function

A function may be represented in several equivalent ways.

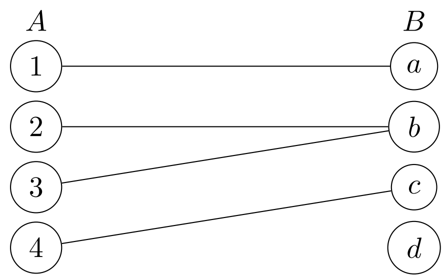
Ordered pairs :

Example:

$$f = \{(1, a), (2, b), (3, b), (4, c)\}$$

Arrow diagram :

Example:



Table

x	1	2	3	4
$f(x)$	a	b	b	c

Formula

Example :

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x + 1.$$

Graph:

If $f : D \to \mathbb{R}$, where $D \subseteq \mathbb{R}$, then the graph of f is

$$G_f = \{(x, f(x)) : x \in D\} \subseteq \mathbb{R}^2.$$

2 Equality of Functions

Definition

Two functions $f : A \to B$ and $g : C \to D$ are equal if and only if:

1. $A = C$;
2. $B = D$;
3. $f(x) = g(x)$ for every $x \in A$.

Example 2

The rules

$$f(x) = (x+1)^2 \quad \text{and} \quad g(x) = x^2 + 2x + 1$$

define the same function if they have the same domain and codomain, because $f(x) = g(x)$ for every input.

However,

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2,$$

and

$$g : [0, \infty) \to \mathbb{R}, \quad g(x) = x^2,$$

are not equal functions because their domains are different.

3 Special Types of Functions

Injective Functions

Definition

A function $f : A \to B$ is *injective* or *one-to-one* if

$$f(x_1) = f(x_2) \implies x_1 = x_2, \qquad \text{for all} \;\; x_1, x_2 \in A.$$

Equivalently,

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

Surjective Functions

Definition

A function $f : A \to B$ is *surjective* or *onto* if

$$f(A) = B.$$

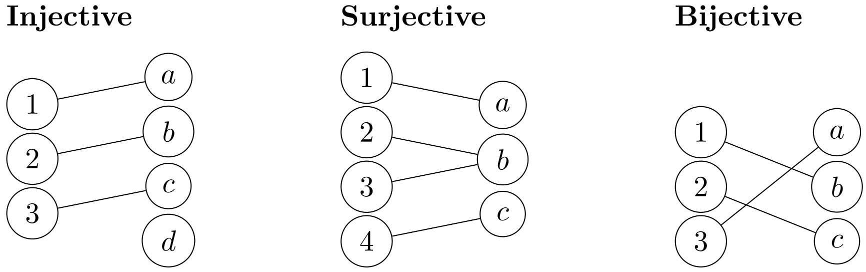
Equivalently, for every $y \in B$, there exists $x \in A$ such that $f(x) = y$.

Bijective Functions

Definition

A function is *bijective* if it is both injective and surjective.

Example 3: Graphical Intepretation



Example 4

Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 2x - 3$.

To prove injectivity, suppose $f(x_1) = f(x_2)$.

Then $2x_1 - 3 = 2x_2 - 3$.

So $x_1 = x_2$.

Thus, f is injective.

To prove surjectivity, let $y \in \mathbb{R}$.

Solving $y = 2x - 3$ gives $x = \frac{y+3}{2} \in \mathbb{R}$.

Therefor $f\left(\frac{y+3}{2}\right) = y$

Thus, f is surjective.

Thus, f is bijective.

Activity 2: Classify the Functions

Determine whether each function is injective, surjective, bijective, or none.

1. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.
2. $g : [0, \infty) \to [0, \infty)$, $g(x) = x^2$.
3. $h : \mathbb{Z} \to \mathbb{Z}$, $h(n) = n + 1$.
4. $p : \mathbb{Z} \to \mathbb{Z}$, $p(n) = 2n$.
5. $q : \mathbb{R} \to (0, \infty)$, $q(x) = e^x$.

Constant and Identity Functions

A *constant function* has the form

$$f(x) = c$$

for every x in its domain.

The *identity function* on a set A is

$$I_A : A \to A, \qquad I_A(x) = x.$$

4 Composition of Functions

Definition

Let $f : A \to B$ and $g : B \to C$. The *composition* of g with f is the function

$$g \circ f : A \to C$$

defined by

$$(g \circ f)(x) = g(f(x)).$$

The function on the right acts first:

$$x \xmapsto{f} f(x) \xmapsto{g} g(f(x)).$$

Example 5

Let

$$f(x) = 2x + 1, \qquad g(x) = x^2.$$

Then

$$(g \circ f)(x) = g(2x + 1) = (2x + 1)^2,$$

while

$$(f \circ g)(x) = f(x^2) = 2x^2 + 1.$$

Therefore, in general,

$$g \circ f \neq f \circ g.$$

Domain of a Composite Function

If the functions are given only by formulas, then

$$\mathrm{Dom}(g \circ f) = \{x \in \mathrm{Dom}(f) : f(x) \in \mathrm{Dom}(g)\}.$$

Example 6

Let

$$f(x) = x - 1, \qquad g(x) = \sqrt{x}.$$

Then

$$(g \circ f)(x) = \sqrt{x - 1},$$

which is defined when $x \geq 1$. Hence

$$\mathrm{Dom}(g \circ f) = [1, \infty).$$

Associativity and Identity

Whenever the compositions are defined,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Also,

$$f \circ I_A = f, \qquad I_B \circ f = f$$

for every $f : A \to B$.

Activity 3

Let

$$f(x) = 3x - 2, \qquad g(x) = \frac{1}{x}, \qquad h(x) = x^2 + 1.$$

Find the formulas and natural domains of:

1. $g \circ f$;
2. $f \circ g$;
3. $h \circ g$;
4. $g \circ h$.

5 Inverse Functions

Definition

Let $f : A \to B$. An *inverse function* of f is a function

$$f^{-1} : B \to A$$

such that

$$f^{-1} \circ f = I_A \qquad \text{and} \qquad f \circ f^{-1} = I_B.$$

Theorem

Theorem

A function has an inverse if and only if it is bijective.

Proof. Suppose f has an inverse. If $f(x_1) = f(x_2)$, then applying f^{-1} gives $x_1 = x_2$, so f is injective. For any $y \in B$,

$$y = f(f^{-1}(y)),$$

so f is surjective.

Conversely, if f is bijective, then for each $y \in B$ there is exactly one $x \in A$ such that $f(x) = y$. Define $f^{-1}(y) = x$. This gives the required inverse function. □

Finding an Inverse Algebraically

To find the inverse of f:

1. write $y = f(x)$;
2. solve for x in terms of y;
3. interchange the symbols x and y;
4. state the domain and codomain carefully.

Example 7

Let
$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 3x - 5.$$
Write
$$y = 3x - 5.$$
Then
$$x = \frac{y+5}{3}.$$
Therefore
$$f^{-1}(x) = \frac{x+5}{3}.$$
Indeed,
$$f^{-1}(f(x)) = \frac{3x-5+5}{3} = x.$$

Remark

The function $f(x) = x^2$ is not injective on $\mathbb{R}$, so it has no inverse as a function $\mathbb{R} \to \mathbb{R}$. However, the restriction
$$f : [0, \infty) \to [0, \infty), \qquad f(x) = x^2,$$
is bijective and has inverse
$$f^{-1}(x) = \sqrt{x}.$$

6 Images and Preimages of Sets

Let $f : A \to B$, $S \subseteq A$, and $T \subseteq B$.

Definition

The *image* of S under f is
$$f(S) = \{f(x) : x \in S\}.$$
The *preimage* of T under f is
$$f^{-1}(T) = \{x \in A : f(x) \in T\}.$$

Remark

The notation $f^{-1}(T)$ denotes a preimage and is meaningful for every function, even when f has no inverse function.

Important Identities

For $S_1, S_2 \subseteq A$ and $T_1, T_2 \subseteq B$:

$$f(S_1 \cup S_2) = f(S_1) \cup f(S_2),$$

$$f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2),$$

with equality in the second formula when f is injective.

Moreover,

$$f^{-1}(T_1 \cup T_2) = f^{-1}(T_1) \cup f^{-1}(T_2),$$
$$f^{-1}(T_1 \cap T_2) = f^{-1}(T_1) \cap f^{-1}(T_2),$$

and

$$f^{-1}(B \setminus T) = A \setminus f^{-1}(T).$$

Example 8

Let

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

If $S = [-2, 1]$, then

$$f(S) = [0, 4].$$

If $T = [1, 9]$, then

$$f^{-1}(T) = [-3, -1] \cup [1, 3].$$

Activity 4: Images and Preimages

Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = 2x - 1$. Find:

1. $f([0, 3])$;
2. $f((-2, 1])$;
3. $f^{-1}([1, 5])$;
4. $f^{-1}((3, \infty))$.

7 Counting Functions Between Finite Sets

Suppose $|A| = m$ and $|B| = n$.

Number of Functions

Each of the m elements of A may be assigned independently to any of the n elements of B. Therefore,

$$\boxed{\text{Number of functions } A \to B = n^m.}$$

Number of Injective Functions

If $m \leq n$, then

$$\boxed{\text{Number of injections } A \to B = n(n-1)\cdots(n-m+1) = \frac{n!}{(n-m)!}.}$$

If $m > n$, there are no injections $A \to B$.

Number of Bijections

If $|A| = |B| = n$, then

$$\boxed{\text{Number of bijections } A \to B = n!.}$$

Activity 5: Counting

Let A have 4 elements and B have 6 elements.

1. How many functions are there from A to B?
2. How many injective functions are there from A to B?
3. Can there be a surjective function from A to B? Explain.

8 Real Functions and Their Graphs

A *real function* in this lesson is a function of the form

$$f : D \to \mathbb{R}, \qquad D \subseteq \mathbb{R}.$$

Its graph is

$$G_f = \{(x, f(x)) : x \in D\}.$$

8.1 The Vertical-Line Test

A set of points in the coordinate plane is the graph of a function of x if and only if every vertical line intersects the graph at most once.

Activity 6: Reading a Graph

Identify the following for the following function, given as a graph. “‘latex

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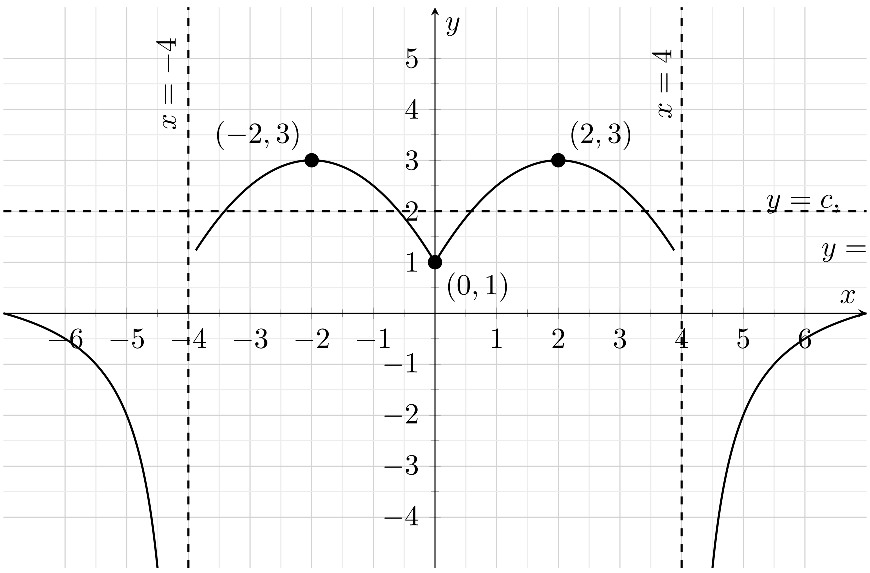

The graph represents the function

[f(x)= $\begin{cases} 1 + \dfrac{3}{x+4}, & x < -4, \\[2ex] -\dfrac{1}{2}(x+2)^2 + 3, & -4 < x \le 0, \\[2ex] -\dfrac{1}{2}(x-2)^2 + 3, & 0 < x < 4, \\[2ex] 1 - \dfrac{3}{x-4}, & x > 4. \end{cases}$]

Activity

The graph of a function (f) is shown above. Identify the following.

- the domain and range;
- x- and y-intercepts;
- intervals where $f(x) > 0$;
- intervals where $f(x) < 0$;
- solutions of $f(x) = c$;
- intervals of increase and decrease;
- maximum and minimum values;
- symmetries and asymptotes.

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9 Special Properties of Functions

Even and Odd Functions

Definition

A function f is *even* if
$$f(-x) = f(x)$$
for every x in its domain. Its graph is symmetric about the y-axis.
A function f is *odd* if
$$f(-x) = -f(x)$$
for every x in its domain. Its graph has rotational symmetry of 180° about the origin.

Increasing and Decreasing Functions

Definition

A function f is increasing on an interval I if
$$x_1 < x_2 \implies f(x_1) \leq f(x_2)$$
for all $x_1, x_2 \in I$. It is strictly increasing if
$$x_1 < x_2 \implies f(x_1) < f(x_2).$$
Decreasing and strictly decreasing functions are defined similarly.

Bounded and Periodic Functions

A function f is bounded above if there exists $M \in \mathbb{R}$ such that $f(x) \leq M$ for every x in the domain. It is bounded below if there exists $m \in \mathbb{R}$ such that $m \leq f(x)$ for every x.

A function f is periodic if there exists $T > 0$ such that
$$f(x+T) = f(x)$$
whenever both sides are defined. The smallest positive such T, when it exists, is called the fundamental period.

10 Transformations of Graphs

Suppose the graph of $y = f(x)$ is known.

New graph	Transformation of $y = f(x)$
$y = f(x) + k$	vertical translation by k units
$y = f(x - h)$	horizontal translation by h units
$y = af(x)$	vertical scaling by factor $\lvert a \rvert$; reflection in the x-axis if $a < 0$
$y = f(bx)$	horizontal scaling by factor $1/\lvert b \rvert$; reflection in the y-axis if $b < 0$
$y = -f(x)$	reflection in the x-axis
$y = f(-x)$	reflection in the y-axis
$y = \lvert f(x) \rvert$	reflect all parts below the x-axis upward
$y = f(\lvert x \rvert)$	keep the right-hand part and reflect it across the y-axis

The general transformed form is

$$y = a\, f\big(b(x - h)\big) + k.$$

Here:

- a controls vertical scaling and reflection in the x-axis;
- b controls horizontal scaling and reflection in the y-axis;
- h controls horizontal translation;
- k controls vertical translation.

Example 9

Starting from $y = x^2$, consider

$$y = -2(x - 3)^2 + 1.$$

The graph is obtained by:

1. shifting $y = x^2$ three units to the right;
2. stretching vertically by a factor of 2;
3. reflecting in the x-axis;
4. shifting one unit upward.

Its vertex is $(3, 1)$ and it opens downward.

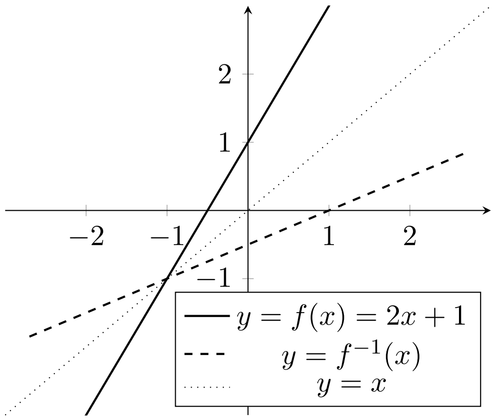
Activity 7: Describe the Transformations

Describe the sequence of transformations that produces each graph from $y = f(x)$.

1. $y = f(x - 4) + 2$;
2. $y = -f(x + 1)$;
3. $y = 3f(2x)$;
4. $y = \frac{1}{2} f\left(\frac{x}{3}\right) - 5$;
5. $y = \lvert f(x) \rvert$;
6. $y = f(\lvert x \rvert)$.

Graph of an Inverse Function

The graph of the inverse relation of f is obtained by reflecting the graph of f in the line $y = x$. If f is bijective, this reflected graph is the graph of the inverse function f^{-1}.



11 Algebraic Operations on Real Functions

Let f and g be real-valued functions. Define

$$(f + g)(x) = f(x) + g(x),$$

$$(f - g)(x) = f(x) - g(x),$$

$$(fg)(x) = f(x)g(x),$$

and

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \qquad g(x) \neq 0.$$

The domain of each new function is the set of inputs for which all required expressions are defined.

Example 10

Let

$$f(x) = \sqrt{x+1}, \qquad g(x) = x - 2.$$

Then

$$\left(\frac{f}{g}\right)(x) = \frac{\sqrt{x+1}}{x-2}.$$

The natural domain is

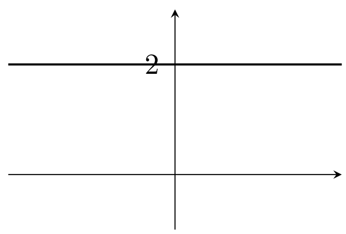
$$[-1, \infty) \setminus \{2\}.$$

12 Catalogue of Standard Real Functions

For each standard function, students should know the formula, natural domain, range, intercepts, symmetry, monotonicity, and important asymptotes.

Constant Function: $f(x) = c$

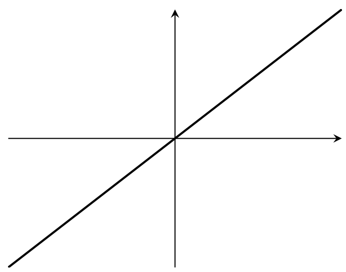
- Domain: $\mathbb{R}$
- Range: $\{c\}$
- Horizontal graph
- Even
- Not injective when the domain has more than one point



Identity and Linear Functions

Identity: $f(x) = x$

- Domain and range: $\mathbb{R}$
- Strictly increasing
- Odd
- Bijective
- $f^{-1} = f$

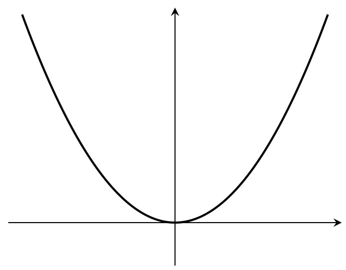


Linear: $f(x) = mx + c$, $m \neq 0$

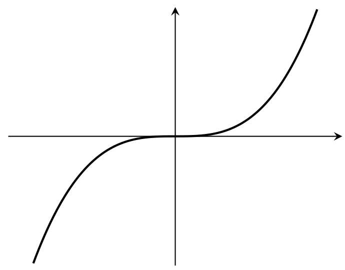
- Domain and range: $\mathbb{R}$
- Increasing if $m > 0$
- Decreasing if $m < 0$
- Bijective as a map $\mathbb{R} \to \mathbb{R}$

Quadratic Function: $f(x) = x^2$

- Domain: $\mathbb{R}$
- Range: $[0, \infty)$
- Even
- Decreasing on $(-\infty, 0]$
- Increasing on $[0, \infty)$
- Not injective on $\mathbb{R}$

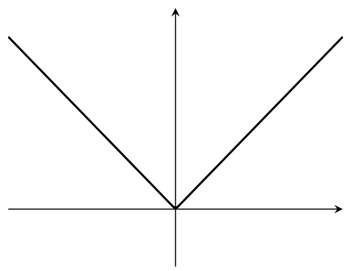


Cubic Function: $f(x) = x^3$



- Domain and range: $\mathbb{R}$
- Odd
- Strictly increasing
- Bijective
- Inverse: $f^{-1}(x) = \sqrt[3]{x}$

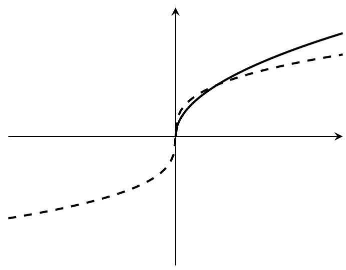
Absolute-Value Function: $f(x) = |x|$



$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

- Domain: $\mathbb{R}$
- Range: $[0, \infty)$
- Even
- Vertex at $(0, 0)$

Square-Root and Cube-Root Functions



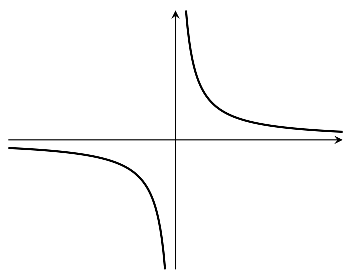
Square root: $f(x) = \sqrt{x}$

- Domain and range: $[0, \infty)$
- Increasing
- Inverse of x^2 restricted to $[0, \infty)$

Cube root: $g(x) = \sqrt[3]{x}$

- Domain and range: $\mathbb{R}$
- Odd and increasing
- Inverse of x^3

Reciprocal Functions



Reciprocal: $f(x) = 1/x$

- Domain and range: $\mathbb{R} \setminus \{0\}$
- Odd
- Asymptotes: $x = 0$ and $y = 0$
- $f^{-1} = f$

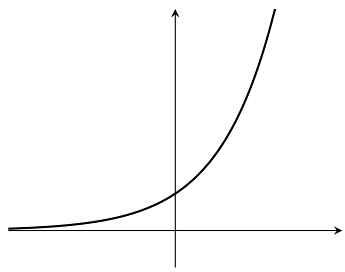
Reciprocal square: $g(x) = 1/x^2$

- Domain: $\mathbb{R} \setminus \{0\}$
- Range: $(0, \infty)$
- Even
- Asymptotes: $x = 0$ and $y = 0$

Exponential Function: $f(x) = a^x$

Assume $a > 0$ and $a \neq 1$.

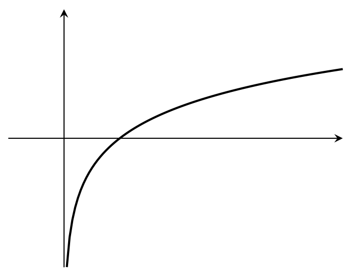
- Domain: $\mathbb{R}$
- Range: $(0, \infty)$
- y-intercept: $(0, 1)$
- Horizontal asymptote: $y = 0$
- Increasing if $a > 1$
- Decreasing if $0 < a < 1$



Logarithmic Function: $f(x) = \log_a x$

Assume $a > 0$ and $a \neq 1$.

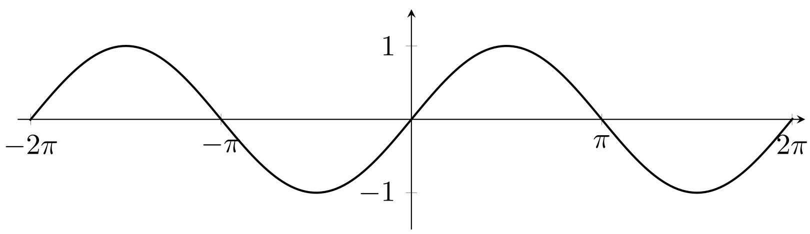
- Domain: $(0, \infty)$
- Range: $\mathbb{R}$
- x-intercept: $(1, 0)$
- Vertical asymptote: $x = 0$
- Inverse of a^x



Trigonometric Functions

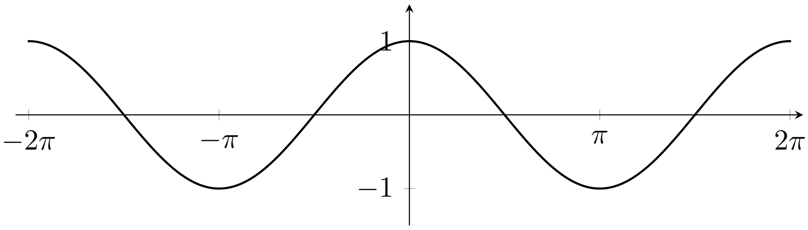
Sine: $f(x) = \sin x$

- Domain: $\mathbb{R}$; range: $[-1, 1]$;
- odd; period 2π;
- zeros at $x = k\pi$, $k \in \mathbb{Z}$.



Cosine: $f(x) = \cos x$

- Domain: $\mathbb{R}$; range: $[-1, 1]$;
- even; period 2π;
- zeros at $x = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$.



Tangent: $f(x) = \tan x$

- Domain:
$$\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi : k \in \mathbb{Z} \right\};$$
- range: $\mathbb{R}$;
- odd; period π;
- vertical asymptotes at $x = \frac{\pi}{2} + k\pi$.

Floor, Ceiling, and Sign Functions

Floor Function

$$f(x) = \lfloor x \rfloor = \text{the greatest integer less than or equal to } x.$$

Its range is $\mathbb{Z}$, and it is discontinuous at every integer.

Ceiling Function

$$f(x) = \lceil x \rceil = \text{the least integer greater than or equal to } x.$$

Its range is $\mathbb{Z}$.

Sign Function

$$\operatorname{sgn}(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0. \end{cases}$$

Piecewise-Defined Functions

A function may use different formulas on different parts of its domain. For example,

$$f(x) = \begin{cases} x+2, & x < 0, \\ x^2, & 0 \le x \le 2, \\ 4, & x > 2. \end{cases}$$

When graphing a piecewise function, open and closed circles indicate whether boundary points are excluded or included.

Activity 8: Standard Functions

For each function below, state its natural domain, range, symmetry, intercepts, monotonicity, and asymptotes where relevant. Then sketch the graph.

1. $f(x) = x^2 - 4$;
2. $g(x) = |x-2| + 1$;
3. $h(x) = \dfrac{1}{x+1}$;
4. $p(x) = 2^x - 3$;
5. $q(x) = \ln(x-2)$;
6. $r(x) = \sqrt{4-x}$.

Important Notations

Notation	**Meaning**
$f : A \to B$	f is a function from A to B
$f(x)$	value of f at x
$f(A)$ or $\mathrm{Ran}(f)$	range of f
$g \circ f$	composition: $(g \circ f)(x) = g(f(x))$
I_A	identity function on A
f^{-1}	inverse function, when it exists
$f(S)$	image of the subset S
$f^{-1}(T)$	preimage of the subset T
G_f	graph of f
$\lfloor x \rfloor$	floor of x
$\lceil x \rceil$	ceiling of x

Important Identities

$$(h \circ g) \circ f = h \circ (g \circ f).$$
$$f \circ I_A = f.$$
$$I_B \circ f = f.$$
$$f^{-1} \circ f = I_A.$$
$$f \circ f^{-1} = I_B.$$

$$f(S_1 \cup S_2) = f(S_1) \cup f(S_2).$$
$$f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2).$$
$$f^{-1}(T_1 \cup T_2) = f^{-1}(T_1) \cup f^{-1}(T_2).$$
$$f^{-1}(T_1 \cap T_2) = f^{-1}(T_1) \cap f^{-1}(T_2).$$
$$f^{-1}(B \setminus T) = A \setminus f^{-1}(T).$$

End-of-Lesson Summary

A function assigns exactly one output to every input.
The domain, codomain, and range must be distinguished carefully.
Injective functions do not repeat outputs, surjective functions reach every element of the codomain, and bijective functions are precisely the invertible functions.
Composition applies functions in sequence and is associative but generally not commutative.
For real functions, graphs reveal domain, range, intercepts, monotonicity, symmetry, and asymptotic behaviour.
Most familiar graphs can be generated from a small catalogue of standard functions by translations, scalings, and reflections.

Review Exercises

1. Give an example of a function that is injective but not surjective.
2. Give an example of a function that is surjective but not injective.
3. Prove that the composition of two injective functions is injective.
4. Prove that the composition of two surjective functions is surjective.
5. Let $f(x) = \dfrac{x-1}{x+2}$. Determine its natural domain and find its inverse.
6. Let $f(x) = x^2$ and $S = [-3, 2]$. Find $f(S)$.
7. Let $f(x) = x^2$ and $T = [4, 16]$. Find $f^{-1}(T)$.
8. How many functions are there from a 5-element set to a 3-element set?
9. How many bijections are there on a 7-element set?
10. Sketch $y = -2|x+1| + 3$ and state its range.
11. Sketch $y = \dfrac{1}{x-2} + 1$ and identify its asymptotes.
12. Describe how the graph of $y = \sqrt{x}$ is transformed to obtain $y = 2\sqrt{3-x} - 1$.